Metropolitan Optical Networks and WDM Ring Topology

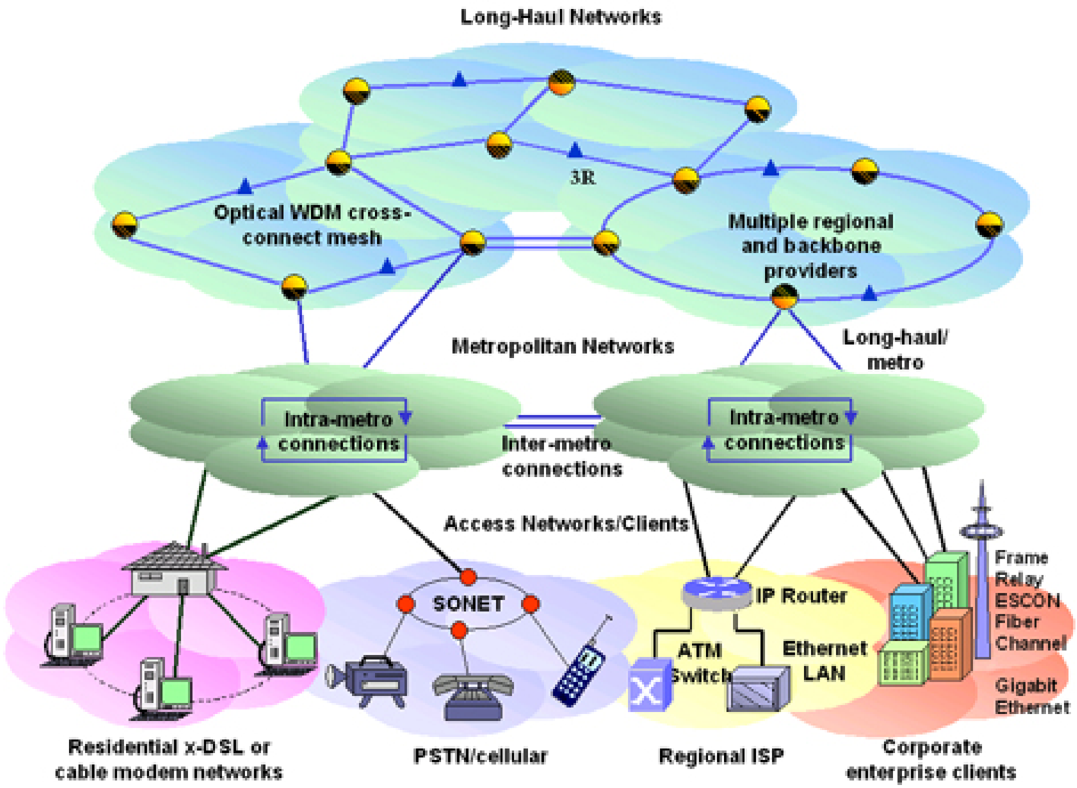
Overview

- General characteristics of metropolitan optical networks (MONs)
  - Key requirements
  - Are they extensions of long-haul optical networks
- Characteristics of basic components
- Introduction of a basic ring topology
- Power budgeting in a dynamic environment
- Should non-linearity be considered
- Migrating to higher bit rates and compensating the dispersion
- New fibers for metro applications and using some un-ideal characteristics of transmitters
- Considering the amplifier gain characteristics
- Inter-channel cross-talk at OXCs
- Implementing new ideas
- Path recovery
- Summary

General characteristics of MONs

- Span distances up to several hundred kilometers
- Typically serve large, concentrated metropolitan areas
- Bridge between long-haul and access networks
- Used to be TDM networks and designed for limited set of traffic types, such as multiplexed voice and private line services
- Rapidly growing capacity demands and increasing variable traffic patterns
- Interconnect a full range of client protocols from enterprise/private customers in access networks to backbone service provider networks
- Driven by two key factors:
  - Customer requirements
  - Technology to support these requirements

A high-level view



A high level view of the overall network hierarchy

“Metropolitan Optical Networks”, White paper, www.sorrentonet.com

Key requirements

- Multi-protocol, multi-service, and sub-rate support
  - DS-0 (64 Kbits/sec), DS-1 (1.5544 Mbits/sec), DS-3 (44.736 Mbits/sec) and 10/100 fast Ethernet, and traditional ATM and frame relay circuits
  - OC-3 (155 Mbits/sec), OC-12 (622 Mbits/sec), OC-48 (2.5 Gbits/sec), OC-192 (10 Gbits/sec)
- Optical transparency
  - Bit rate and format independent. WDM might be the key technology
- Scalability
  - Topological flexibility is required
  - WDM-based rings by using re-configurable OADM are proposed
  - Mesh network topology is another alternative
- Multiple survivability options
  - Protection: dedicated/shared protection, pre-provisioned recovery
  - Restoration: dynamic signal recovery
- Low cost

Extensions of long-haul networks?

	Long-haul	Metro
Transmission distance	Hundreds to thousands of km	Typically up to 100 km
Traffic channels	2.5 Gbps, 10 Gbps, and 40 Gbps	Highly diverse, SONET-like channels coexist with Fiber and Gigabit Ethernet channels
Fiber topology	Linear	Ring or mesh
Traffic topology	Linear	Hub-and-spoke
Traffic protection	Implemented by OXC, DCC, or upper layer protocol. Transmission system itself does not provide protection	Transmission system must provide protection or support client layer protection mechanisms
Migration	Limited to additional channels	Addition of both capacity and new ring sites
Cost	Low cost is an advantage	Must be low enough to compete with alternative technologies and the provision of additional fiber
Footprint	Small size is an advantage	Small size is crucial

Basic components

- Transmitters
  - Externally or directly modulated lasers
- Wavelength add/drop Multiplexers
  - Fixed or re-configurable
- Switches and optical cross-connects (OXC)
  - Fiber switching
  - Wavelength switching
  - Wavelength conversion
- Optical amplifiers
  - Lower noise and flatter, stable gain is desired
- Fibers
- Receivers
  - Photo-detectors: PIN, APD (avalanche photodiode)

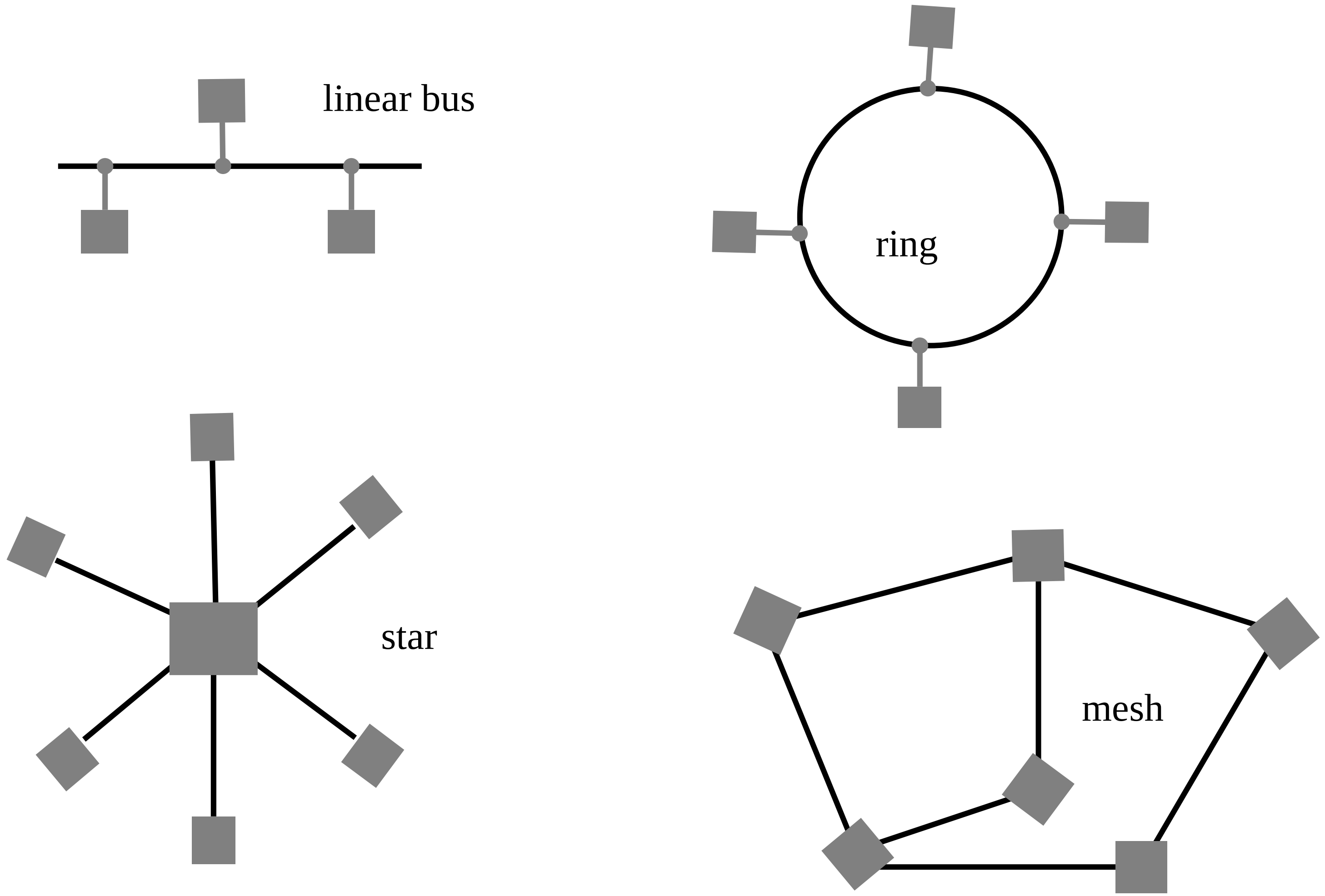
Designing the optical layer

- Span design
  - Should consider loss, OSNR, dispersion, and nonlinear effects
  - OXCs and OADMs dynamically change the signal path and increase the complexity of the design
- Wavelength routing plans
  - Each wavelength can follow a different path which might be envisioned as a virtual topology, such as point-to point, mesh or ring system
    - Wavelength path: the signal uses the same wavelength from the source to the sink
    - Virtual wavelength path: the signal can travel on different wavelengths for an end-to-end connection

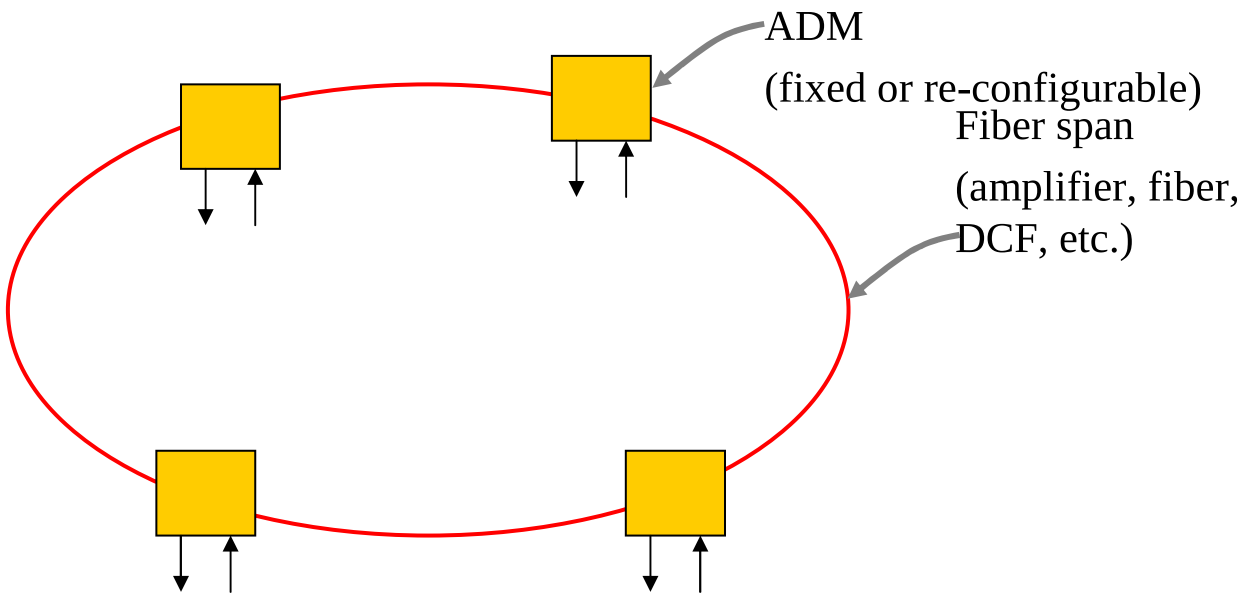
Designing the optical layer

- Restoration
  - Performing restoration at the optical layer provides cost saving
  - Restoration types:
    - Protection: dedicated/shared protection, pre-provisioned recovery
    - Restoration: dynamic signal recovery
- Network management
  - Managing the overall network architecture requires an intelligent network management system (NMS)
  - NMS should be able to monitor signal performance of each wavelength

Classical options



Starting with a basic ring network



- Consider a typical WDM metro ring at four OADMs, as shown in the above figure.

Power level management

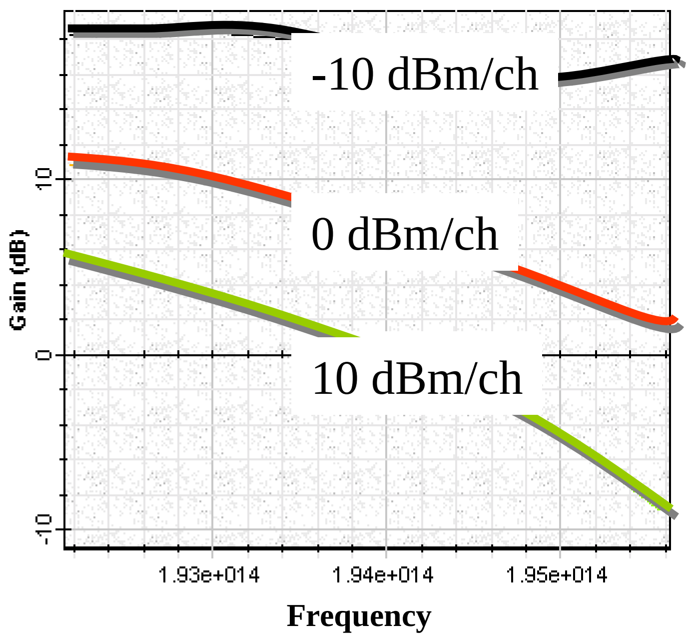
Goals of this example:

- *To discover the power related issues in metro networks*
- *To show the power compensation/management by using amplifiers*
- *To create a basic ring network*

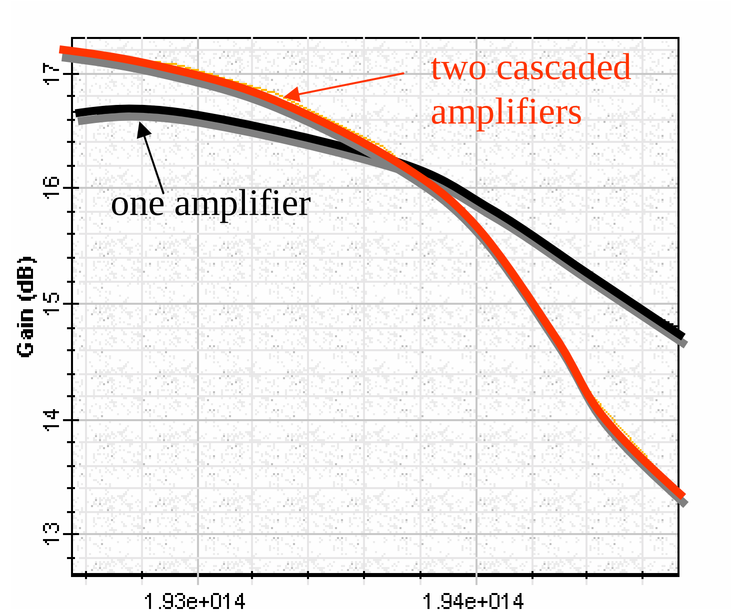
- We need to consider all of the optical power variations resulting from:
  - Wavelength dependent loss in the fibers,
  - Wavelength dependent gain in the amplifiers,
  - Channel-to-channel insertion loss variation in the multiplexers/demultiplexers.

Optical amplifier

- The gain and the gain tilt of the optical amplifier change with the input power



•Gain and gain tilt dependence on input power



•Effect of cascading amplifiers

Sidney Shiba et. al., “Optical Power Level Management in Metro Networks”, NFOEC’01, 2001.

OADMs and OXCs

OADM:

- Optical add/drop Multiplexers (OADMs) introduce insertion loss
- Loss of ADM includes two parts; Deterministic and random
- Loss varies with the wavelength

OXC:

- Variation of optical switch types and different path lengths in the network will result in a random loss
- Difficult to predict

Loss compensation

Limiting factors:

- Nonlinear effects
- Receiver sensitivity
- Input and output power level of amplifiers
- Should consider dynamic range, which is typically decided by the gain tilt and noise figure of the amplifier
- Losses in the fiber, OADMs, OXCs

Typical loss values

Symbol	Description	Loss value
L_{mon}	Tap loss for monitoring	0.2 dB
L_{conn}	Loss of a connector	0.25 dB
L_{splice}	Loss of a splice	0.15 dB
L_{span}	Loss of the fiber span	0.25 dB/km
L_{mux}	Loss of a multiplexer	4 dB
L_{demux}	Loss of a demultoplexer	4 dB
L_{foadm}	Loss of a fixed OADM	2 dB
L_{roadm}	Loss of a re-configurable OADM	10 dB

Symbol	Description	Optical Power/Receiver Sensitivity
$P_{T-DM\ @\ 2.5}$	2.5 Gbps/DM output power	3 dBm
$P_{T-EM\ @\ 10}$	10 Gbps/EM output power	0 dBm
$P_{R-sen\ @\ 2.5}$	2.5 Gbps pin diode receiver sensitivity	-23 dBm
$P_{R-sen\ @\ 10}$	10 Gbps pin diode receiver sensitivity	-16 dBm

$$L_{allowable} = P_T - P_{R-sens}$$

$$L_{node} = 2L_{splice} + 6L_{conn} + L_{f/r\ oadm}$$

Gerlas van den Hoven et. al., “Seeking Simple Migration Paths”, WDM Solutions, October 2001.

Typical values for amplifiers

	Units	Metro target	Long-haul EDFA	Metro EDFA	EDWA	SOA	LOA
Gain	dB	10 to 20	>20	10 to 20	10	10 to 20	10 to 20
Power	dBm	10 to 15	>20	15	7	10	10
Noise figure	dB	6	<5	6	6	8	<8
Polarization-dependent gain	dB	0.5	0.1	0.3	0.3	1	<1
Gain flatness	dB	1	5 to 10*	5 to 10*	5 to 10*	1	<1
Intersymbol interference		Low	Low	Low	Low	High	Low
WDM crosstalk		Low	Low	Low	Low	High	Low
Switching transients		Low	High**	High**	High**	High**	Low
Power consumption	W	5	30	3 to 7	5	5	5

$$G_{required} = -\left(P_{channel} - N_{span}L_{span} - N_{span}L_{node} - P_{R-sens} - LM\right)$$

Gerlas van den Hoven et. al., “Seeking Simple Migration Paths”, WDM Solutions, October 2001.

Example

- Without amplification, the allowable loss is given by

$$L_{allowable} = 3 - (-23) = 26\ dB$$

If we consider that the carriers require metro rings with circumferences of up to 600 km and 20 nodes, we conclude that *this cannot be achieved without amplifiers*.

- Total node loss (including two splices and six connectors) is given by

$$L_{node} = 2 \times 0.15 + 6 \times 0.25 + 2 = 3.8\ dB$$

- Span loss will be $L_{span50km} = 50 \times 0.25 = 12.5\ dB$
- Total loss will be $L_{span50km} = 12.5 \times 4 + 3.8 \times 4 = 65.2\ dB$

Design options

In terms of power level control

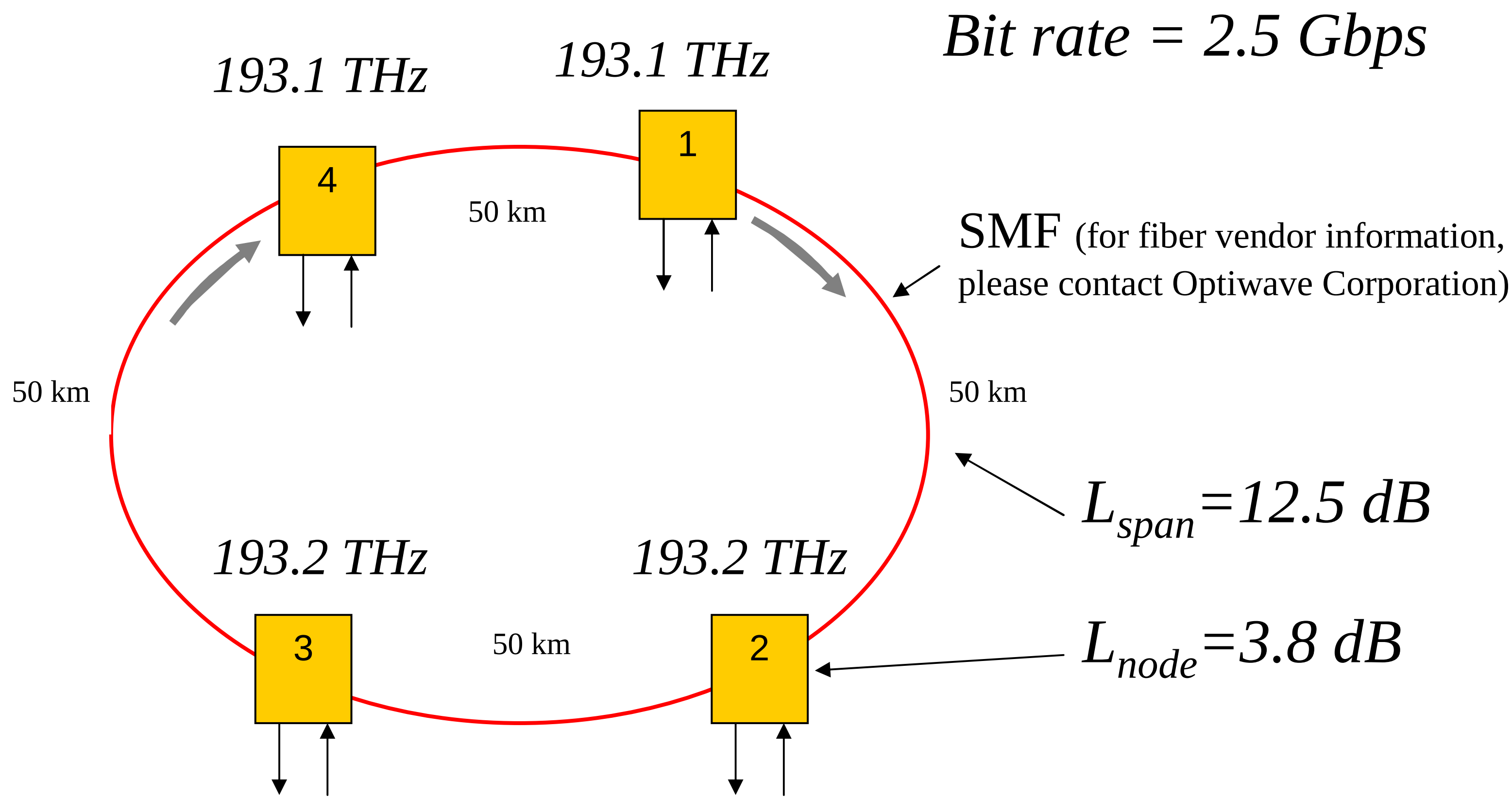
- Design without amplification
  – Requires higher launch power which may trigger the fiber nonlinearity
- Design with amplification

  Two options:

  – “Lumped” amplification

  – Distributed “balanced” amplification

No amplifier option



$P_{required}$=65.2-23-3=39.2 dBm/channel

•At this power level, we need to consider fiber nonlinearity.

Self phase modulation

- SPM effects are negligible when $P_0 < \alpha/\gamma$
- For the fiber we used $\gamma \approx 1.5\, W^{-1} km^{-1}$
- SPM effects can be negligible when the pick power is below *166 mW* or *18 dBm average power*
- If you use N_A amplifiers along the link, the criteria becomes $P_0 < \alpha/(\gamma N_A)$. If you use two amplifiers along the link, maximum allowable power before the nonlinearity becomes effective drops by 3 dB.
- Using dispersion management by using DCF can reduce SPM

Cross phase modulation

- As a rough estimate, the channel power is restricted with $P_{ch} < \alpha / \left[\gamma \left(2N_{ch} - 1 \right) \right]$

  Here, N_{ch} is the number of channels
- For a two channel system limiting power is around *56 mW (17.5 dBm)*. For a *10* wavelength system, the limit is around *10 mW (10 dBm)*
- Separation between channels also affects the XPM
- An increase in the separation will decrease the penalty which originates from the XPM

Four wave mixing

- FWM efficiency depends on signal power, channel spacing, and dispersion
- If the GVD of the fiber is relatively high $|\beta_2| > 5\ ps^2 / km$, the FWM efficiency factor almost vanishes for a typical channel spacing of *50 GHz* or higher
- If the channel is close to zero dispersion wavelength of the fiber, considerably high power can be transferred to FWM components.
- To reduce the effect of FWM to the system performance, you can use either uneven channel spacing or use dispersion-management technique

Stimulated Raman scattering

- The Raman threshold for a single channel system is given by

$$P_{th} \approx \frac{16A_{eff}}{g_R L_{eff}}$$

where $L_{eff} \approx 1/\alpha$ for long fibers

- It is also a function of the number of the channels and the channel power
- For a single channel system, it is around *500 mW* near *1.55 micrometer* if $g_R = 1\times10^{-13}\ m/W$
- For a 20 channel system P_{th} exceeds 10 mW
- It is around 1 mW for a 70 channel system
- It has little impact on system performance

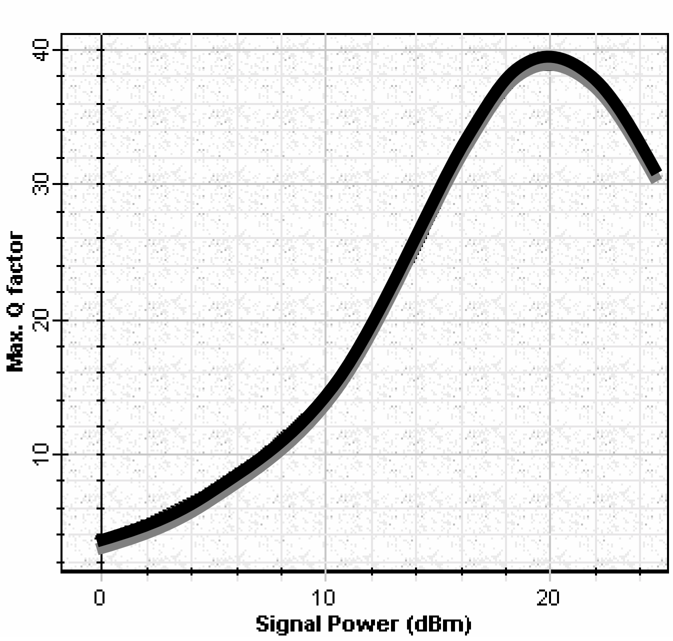
Stimulated Brillion scattering

- Threshold level depends on source line-width, effective core area, and effective fiber length.

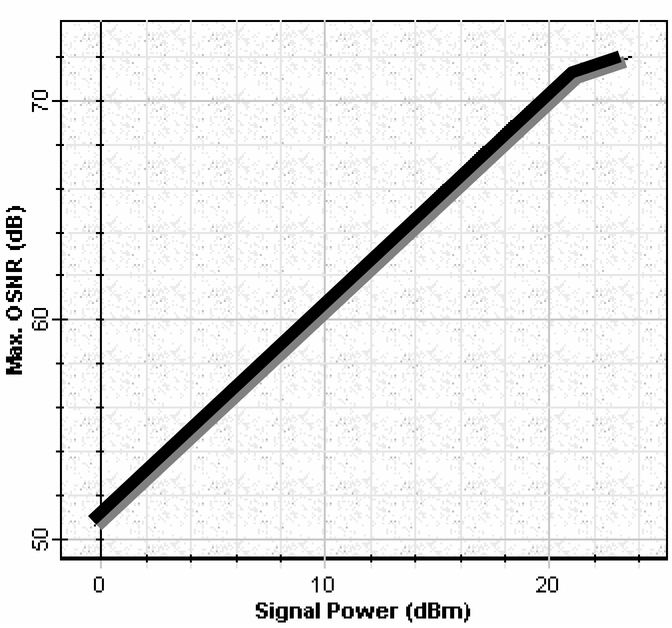
$$P_{th} \approx {21A_{eff}} \big/ {g_B L_{eff}}$$

- Typical value for g_B is around $5\times10^{-11}\ m/W$
- The threshold value also depends on modulation format and duration of pulse.
- Some values:
  - 9 dBm for CW light
  - 12 dBm for externally modulated transmitter
  - >18 dBm for externally modulated transmitter with source wavelength dither
- Generally speaking, SBS has little effect on system performance

System performance



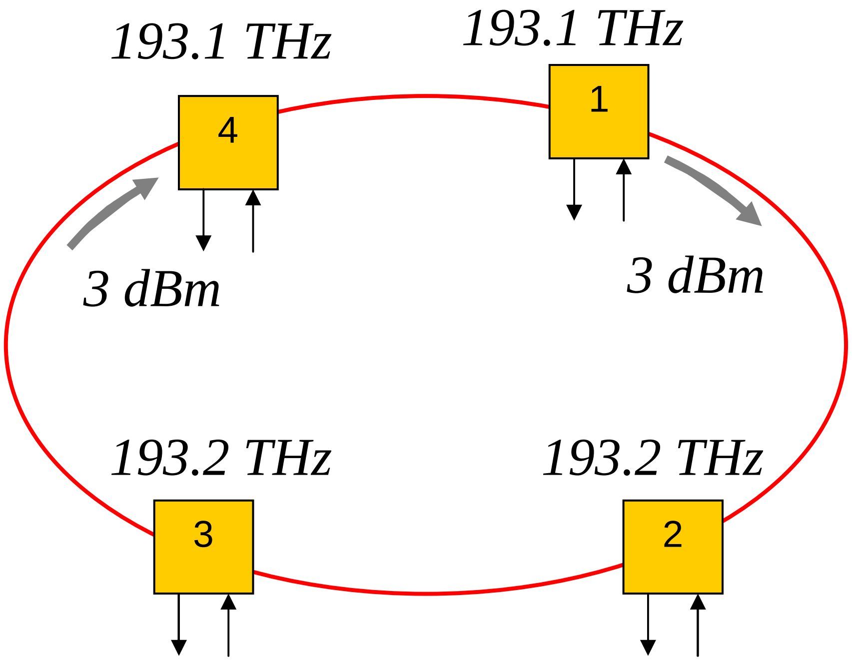
Q factor verses launch power



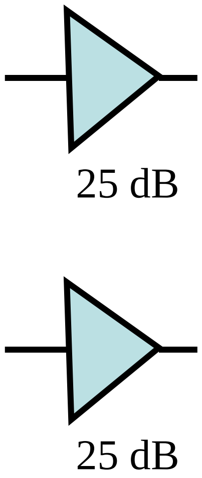
SNR verses launch power

Design with amplifiers

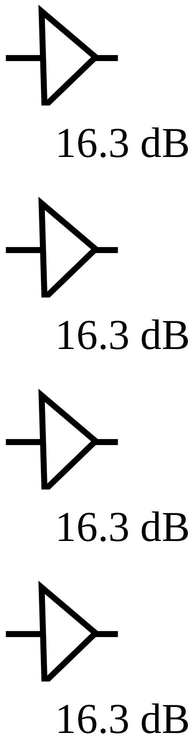
Bit rate = 2.5 Gbps



Alternative 1

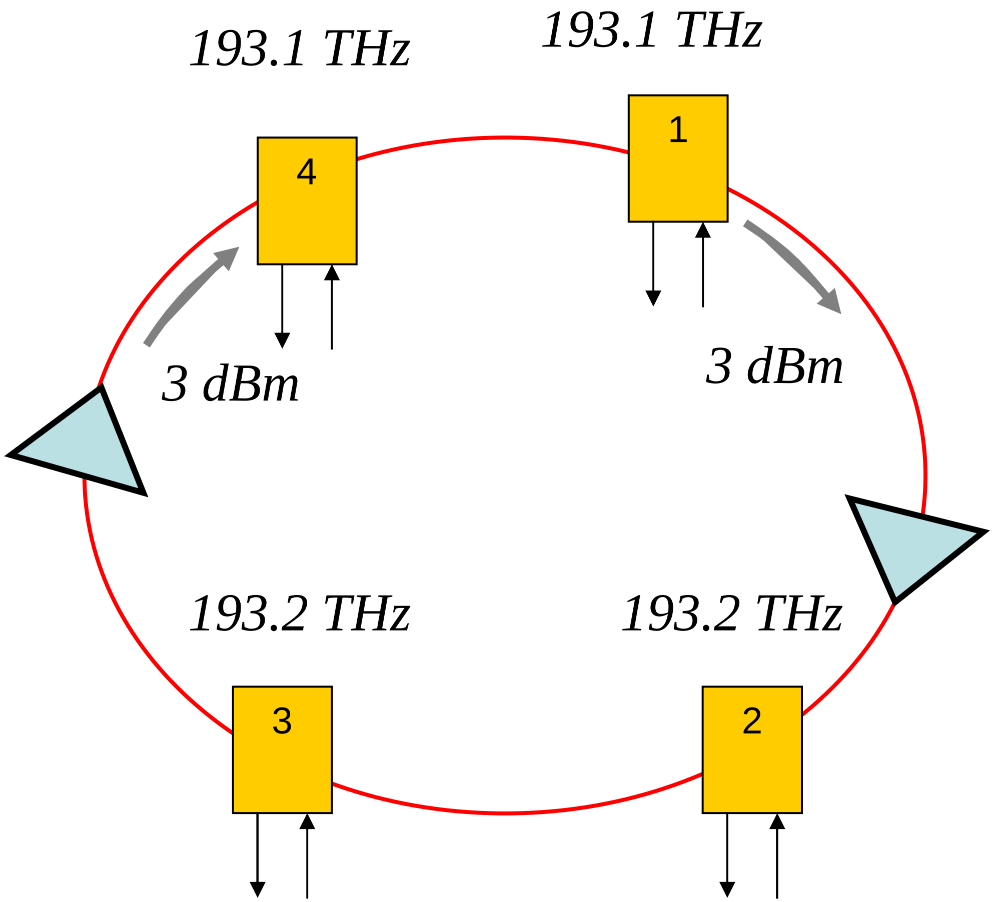


Alternative 2



$$G_{required} = -\left(3 - 4\times 0.25\times 50 - 4\times 3.8 - (-23) - 3\right) = 42.2\ dB$$

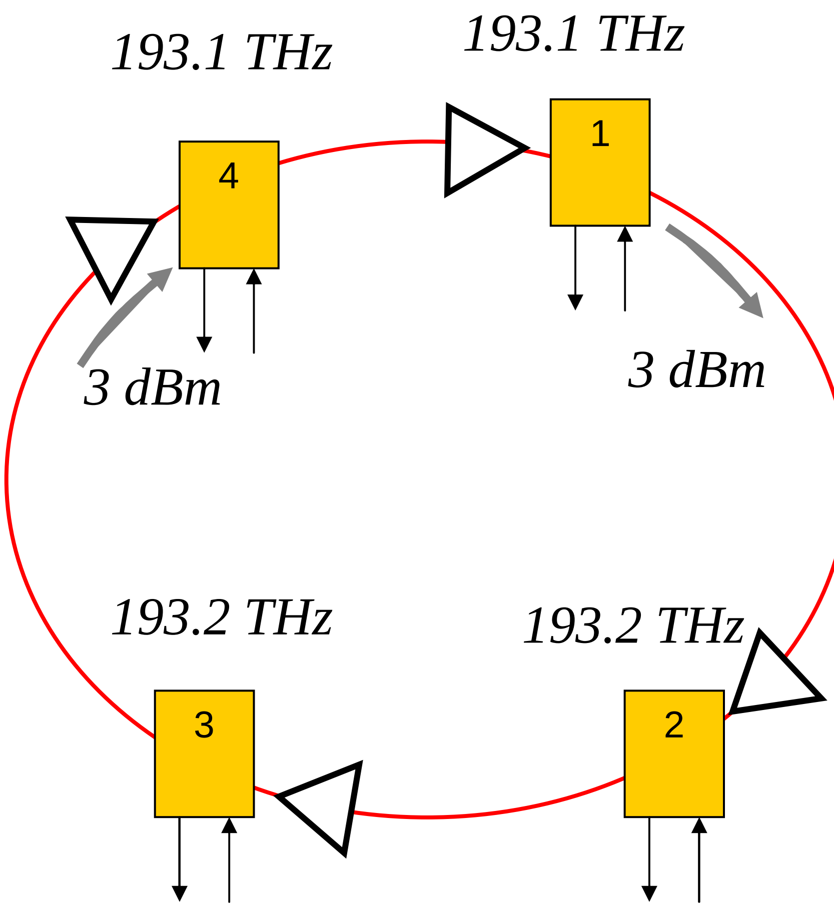
Alternative 1: "Lumped" amplification



- 25 dB is higher than the gain target value for metro EDFA or other types of amplifiers (see table)
- Not topologically flexible (does not allow addition of new node)
- Nonlinearity might be an issue to consider

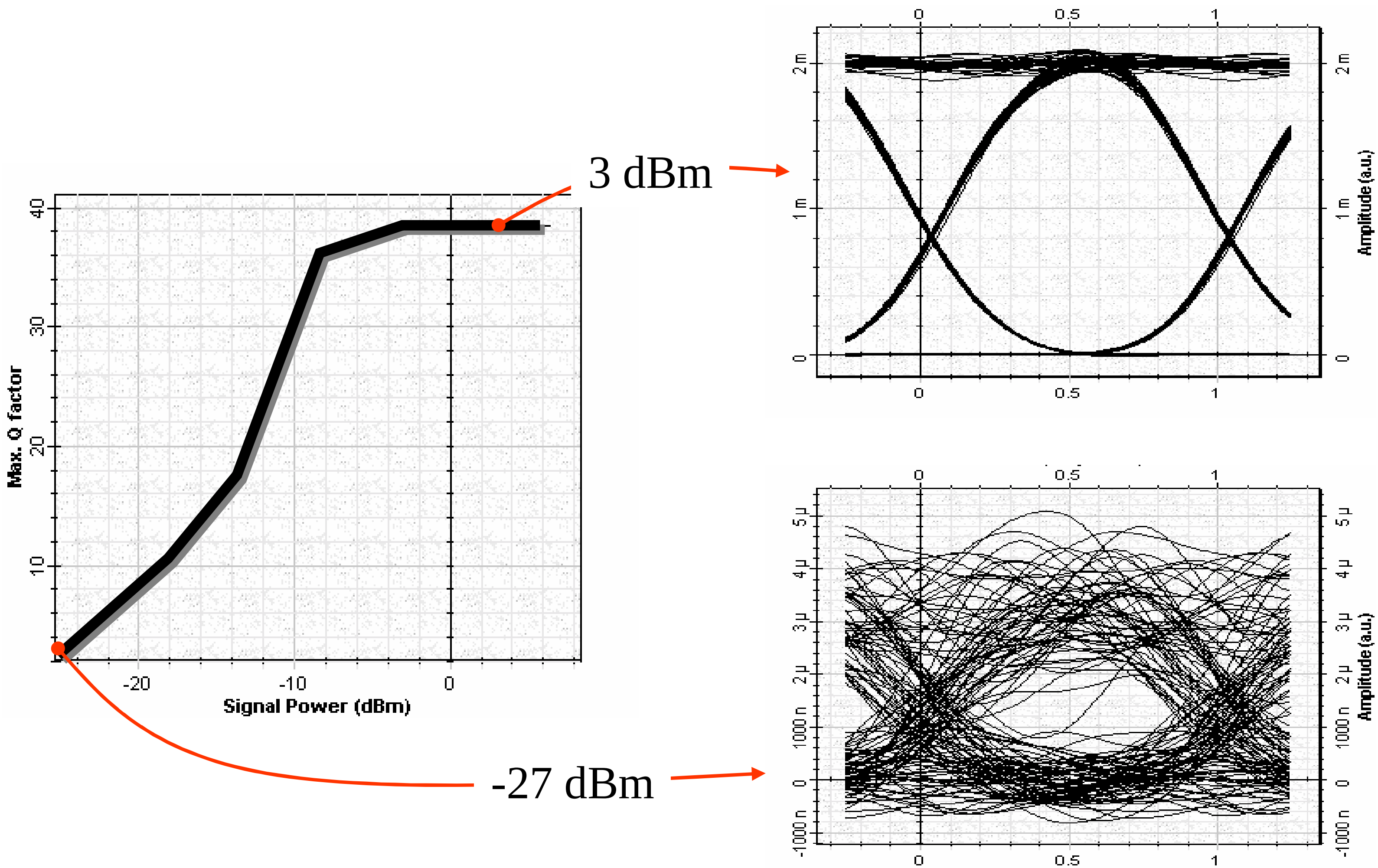
	Units	Metro target
Gain	dB	10 to 20

Alternative 2: one amplifier per node



- Each amplifier will match the loss of one node and the span preceding it
- Brings the *topological flexibility* (a new node can be added easily)
- *Nonlinearity* is *not* an issue
- Requires *low cost-compact* optical amplifiers

Simulation results



Migrating to 10 Gbps

Goals of this example:

- *To show the power budgeting for 10 Gbps*
- *To show the effect of group velocity dispersion at 10 Gbps*

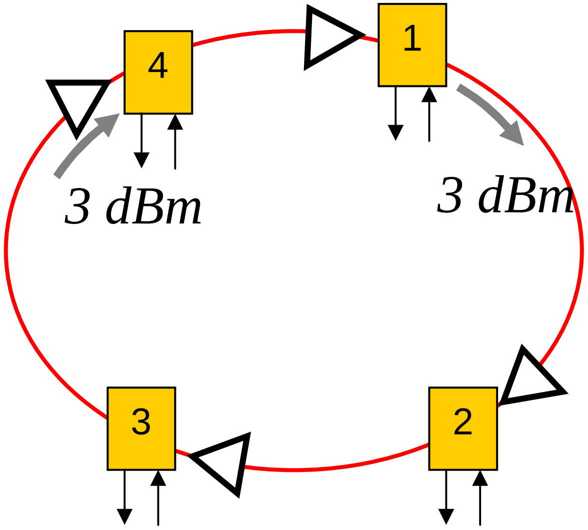
- At 10 Gbps transmission rate, we also need to consider the following additional effects:
  - Reduction of the receiver sensitivity from *–23 dBm* to *–16 dBm.*
  - Group velocity dispersion (GVD).

Loss compensation

- The receiver sensitivity is *–16 dBm*
- Allowable loss is *3- (-16) = 13 db*
- If we assume a power of *3 db* per channel,

$$G_{required} = -\left(3 - 4\times 0.25\times 50 - 4\times 3.8 - (-16) - 3\right) = 49.2\ dB$$

- Better to use unity gain segment approach for loss compensation
- Migration is easily done in terms of power budgeting since the total loss is not changed, even though the allowable loss is changed

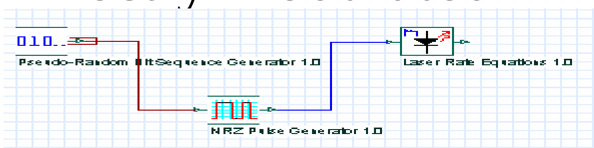


Optiwave

GVD limited transmission distance

- Directly modulated DFB lasers

$$L < \frac{1}{4B|D|\sigma_\lambda}$$



root-mean-square spectral width, a typical value is around *0.15 nm*

For *D=16 ps/(km-nm)*

at *2.5 Gbps*, $L \approx 42$ *km*

- Externally modulated source

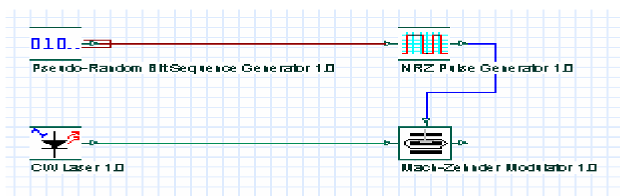
$$L < \frac{2\pi c}{16|D|\lambda^2 B^2}$$

For *D=16 ps/(km-nm)*

at *2.5 Gbps*, $L \approx 500$ *km*

at *10 Gbps*, $L \approx 30$ *km*

Dispersion compensation is required

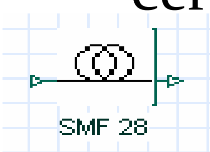


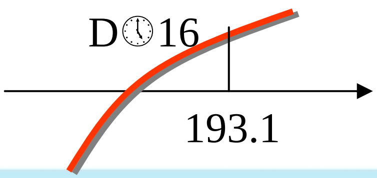
•G. P. Agrawal, Applications of nonlinear fiber optics, Academic Press, 2001.

•R. Ramaswami and K. N. Sivarajan, Optical Networks: A practical Perspective, Morgan Kaufmann, 1998.

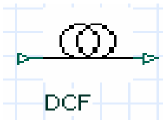
Dispersion compensation

- Total dispersion per *50 km* span is around *16 ×50=800 ps/nm*
- Requires *8 km* of dispersion compensating fiber (DCF) with *D=-100 ps/nm/km* for each fiber span
- Extra amplification is required to compensate the loss in DCF
- Two possible configurations:
  - “Lumped” compensation. After each SMF, one DCF is inserted
  - “Per span” compensation. Dispersion is compensated at certain points in the network





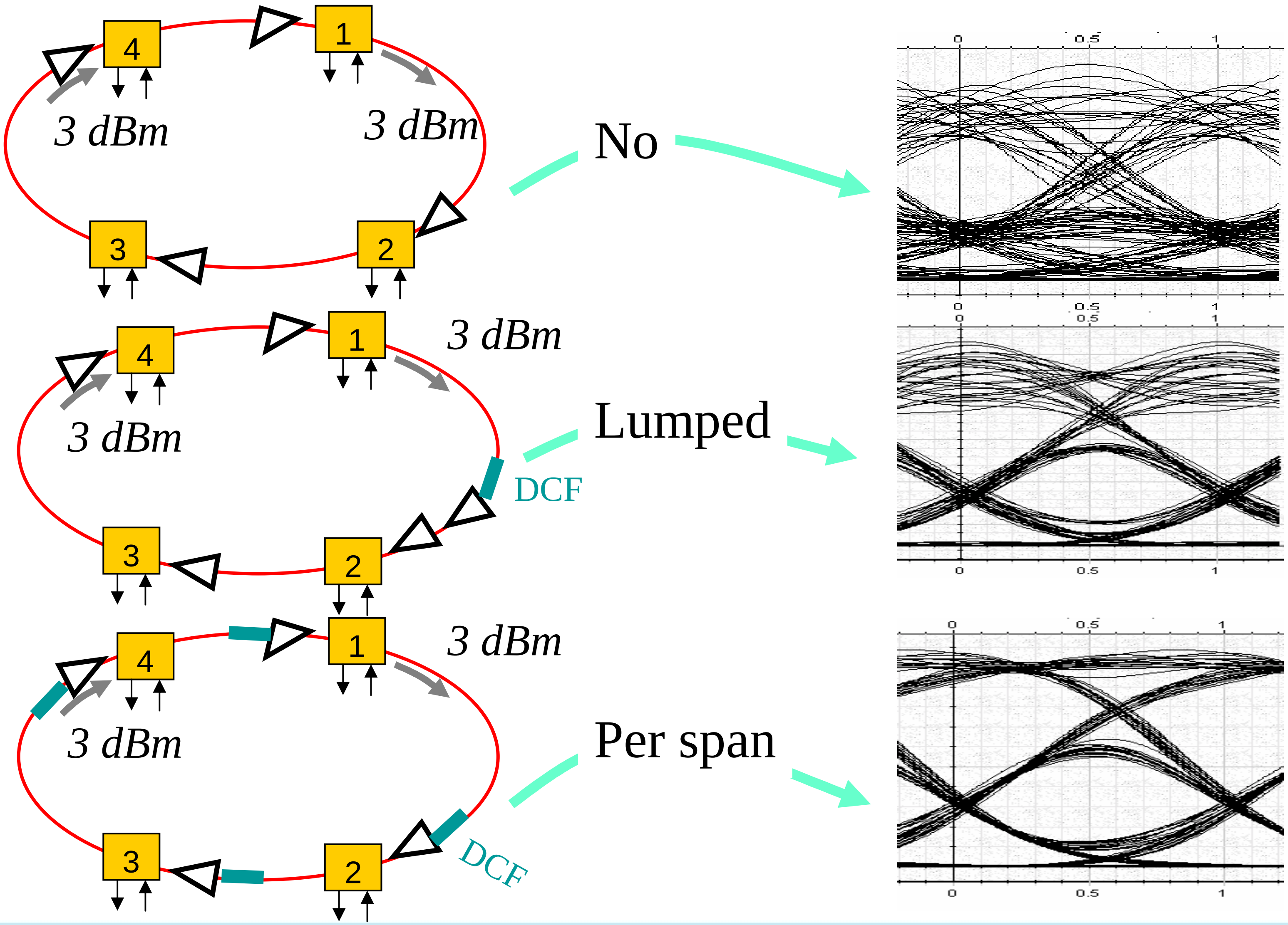
D 16



D -100

- *consider scalability requirement and dynamic nature of the metro network*

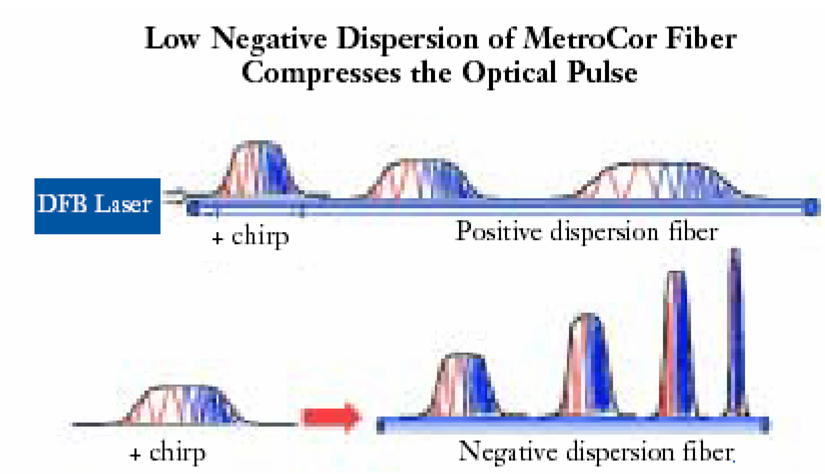
Simulation results

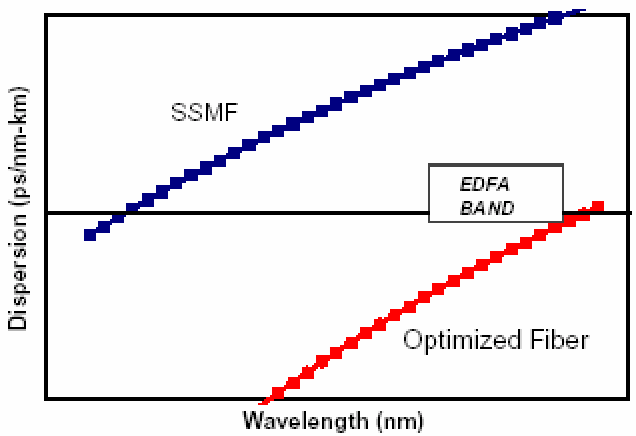


Considering a different fiber

Goals of this example:

- *To compare the performance of two fibers for metro applications*
- *To show the effect of source chirp on performance*

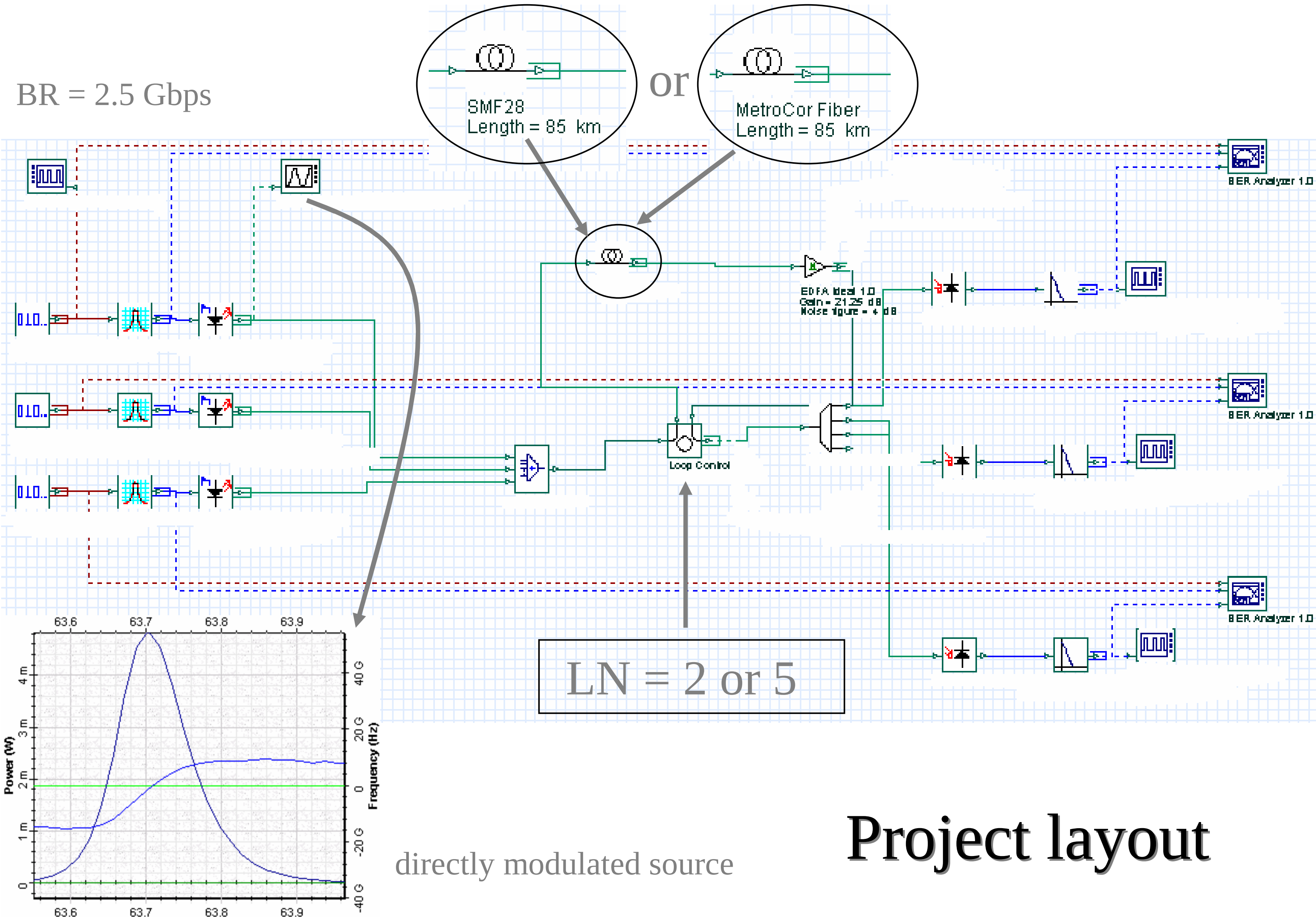




Standard SMF vs Fiber 2

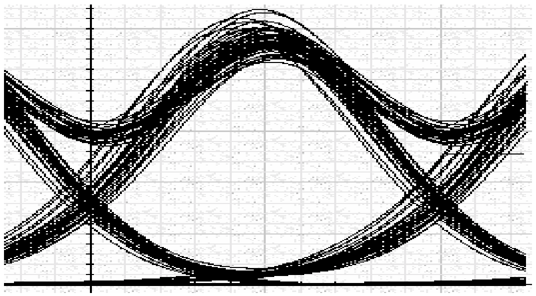
•Chris Kennedy et. al., “The perforance of negative dispersion fiber at 10 GBps and significance of externally and directly modulated lasers”, NFOEC’01, 2001.

•David Culverhouse et. al., “Corning MetroCore Fiber and its Application in Metropolitan Networks”, Corning’s White Paper WP5078.

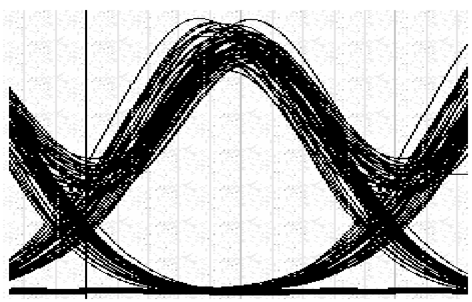


Project layout

Simulation results



•Eye diagram of Channel 1 at *1557.36 nm* after *170 km* of propagation in *SMF* where total accumulated dispersion is *~3000 ps/nm*



•Eye diagram of Channel 1 at *1557.36 nm* after *425 km* of propagation in *Fiber 2* where total accumulated dispersion is *~3000 ps/nm*

• *longer distance and better performance*

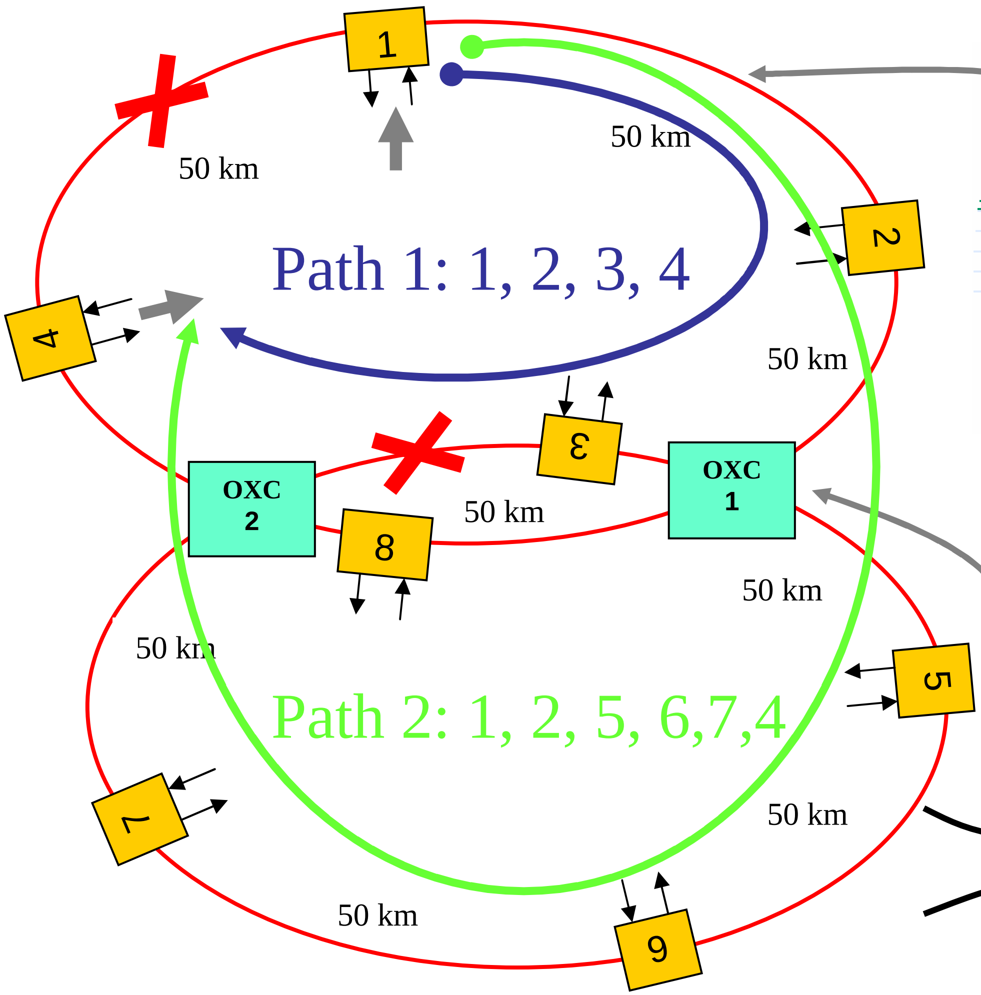
•Chris Kennedy et. al., “The perforance of negative dispersion fiber at 10 GBps and significance of externally and directly modulated lasers”, NFOEC’01, 2001.

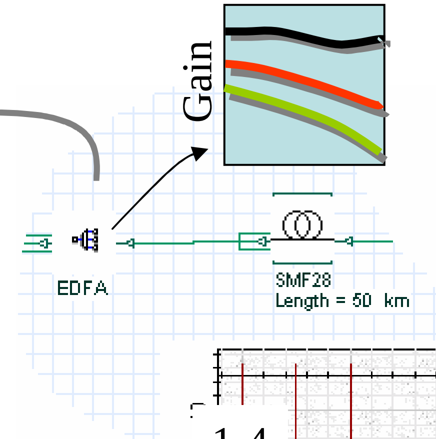
•David Culverhouse et. al., “Corning MetroCore Fiber and its Application in Metropolitan Networks”, Corning’s White Paper WP5078.

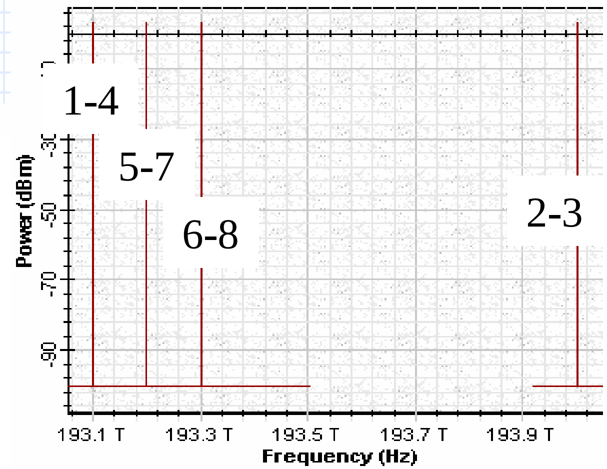
Considering a non-ideal amplifier

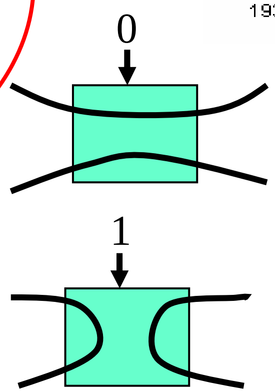
Goals of this example:

- *To discover the effect of unideal amplifier gain to network performance*
- *To design a more complicated project*
  - *Fiber cut*
  - *OXC*
  - *Mesh like network with two interconnected rings*
- *To show one optical layer restoration scheme*



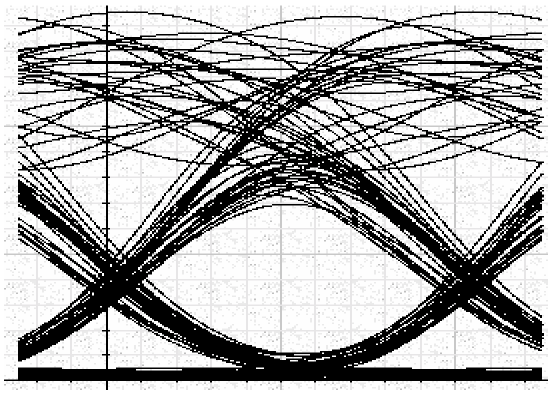






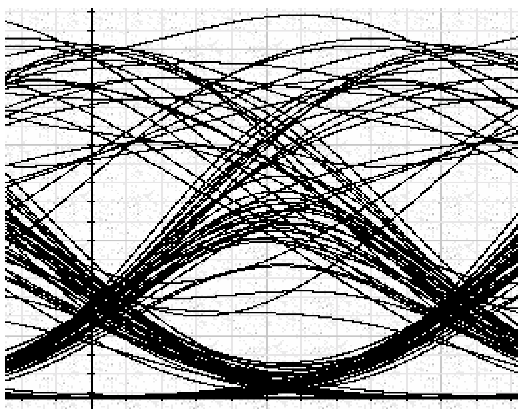
Design layout

Simulation results



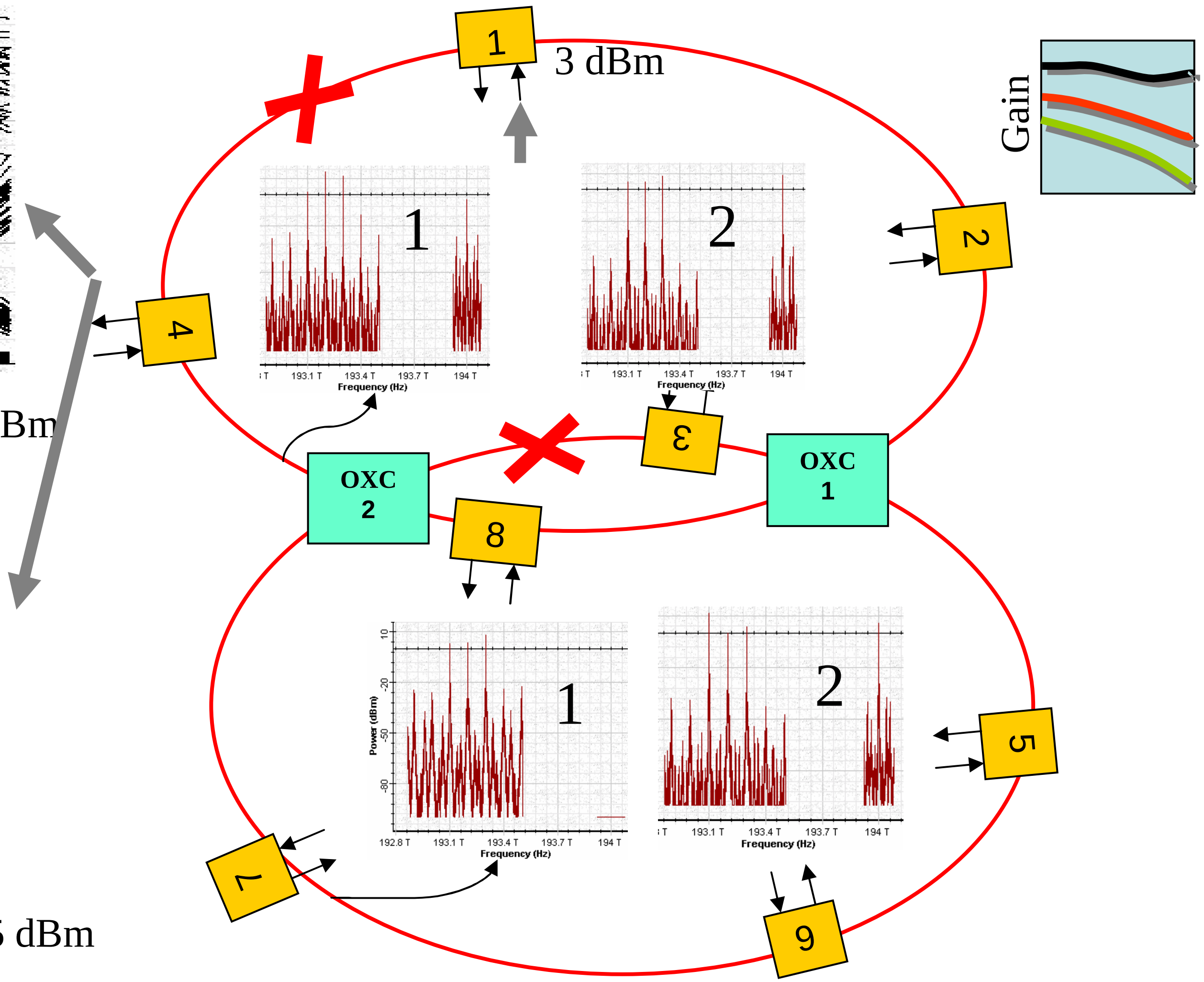
Received signal power is 9.8 dBm

SNR=43 dB



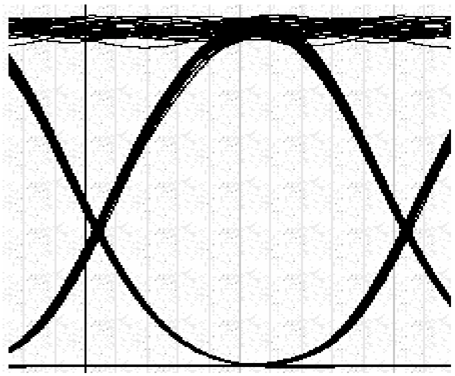
Received signal power is 15.5 dBm

SNR=40 dB



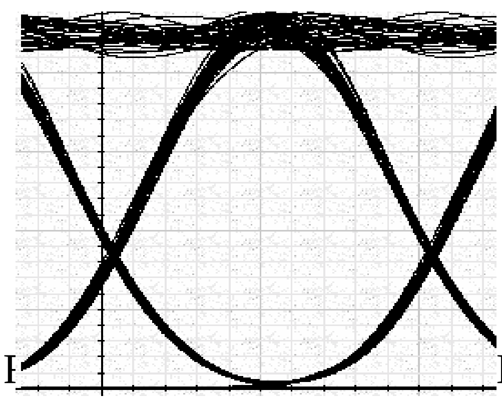
Gain flattening and control is required

Simulation results



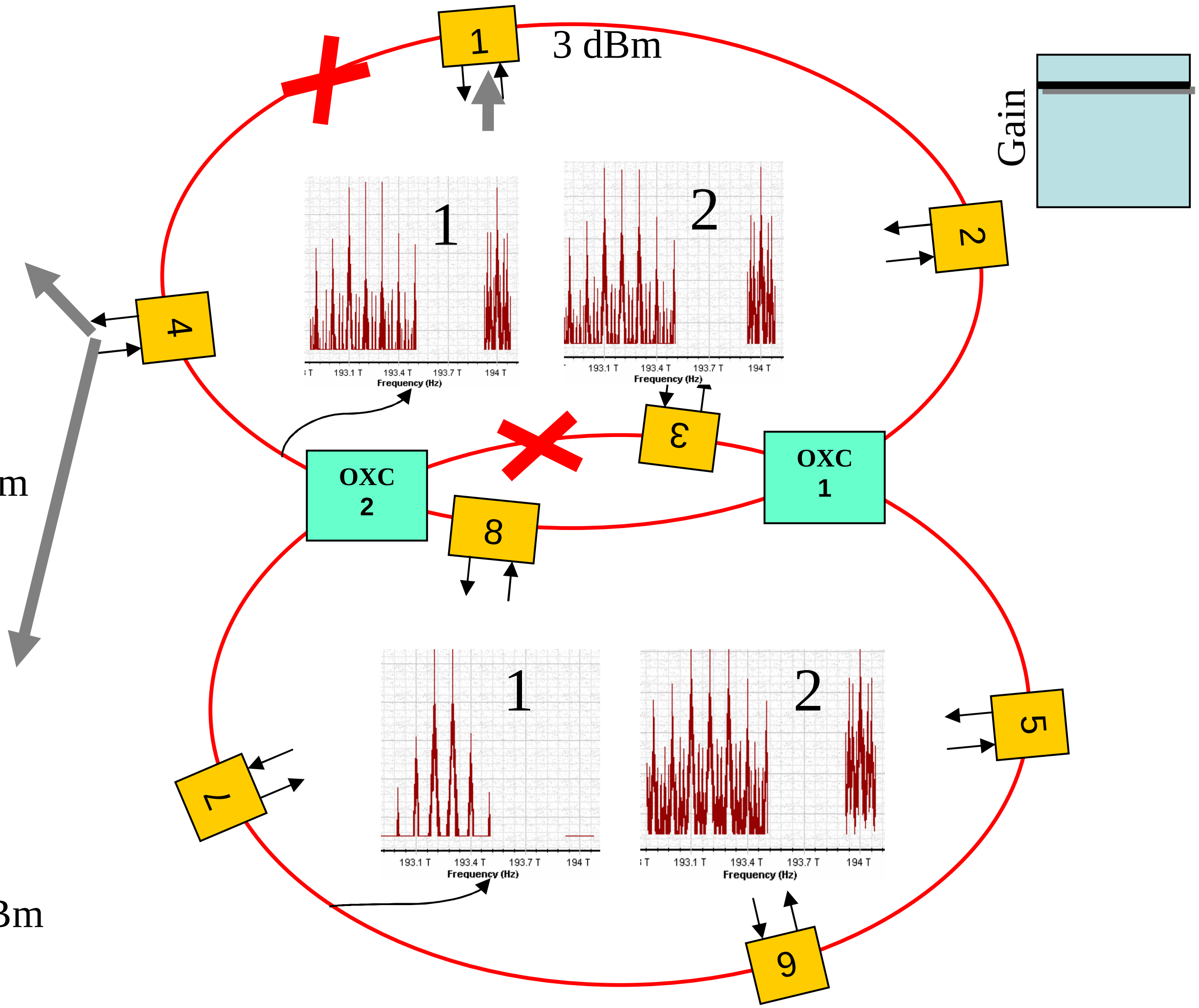
Received signal power is 3 dBm

SNR=34 dB



I Bm

SNR=32 dB

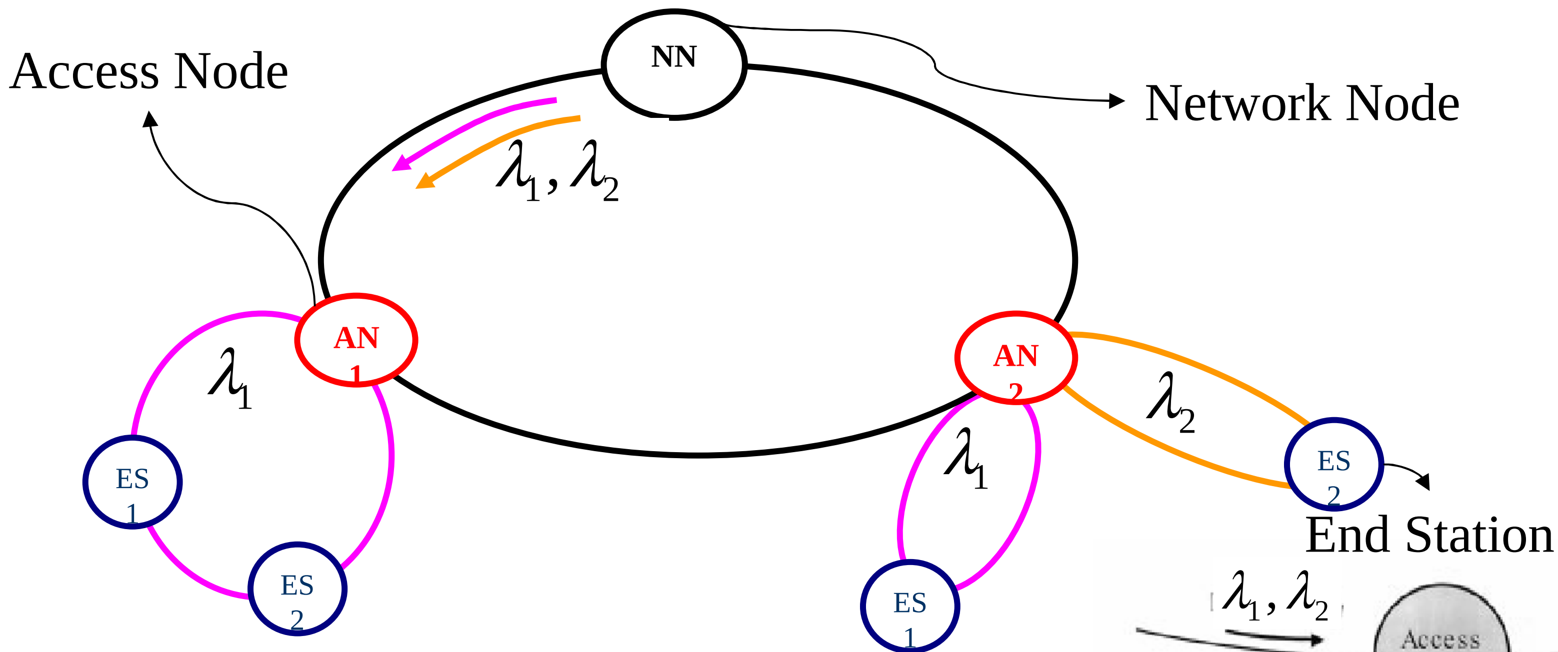


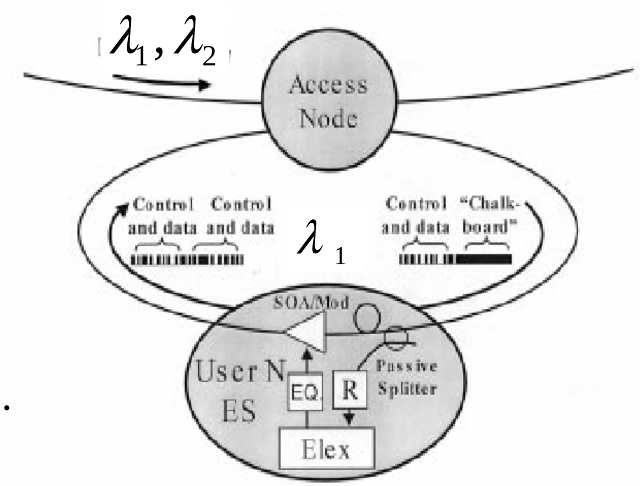
Wavelength independent subscriber

Goals of this example:

- *Realization of a WDM ring network using wavelength independent subscriber equipment to share bandwidth*
- *Effect of source type to performance*
- *Effect of bit rate to performance*
- *Effect of modulation type to performance*

Network layout

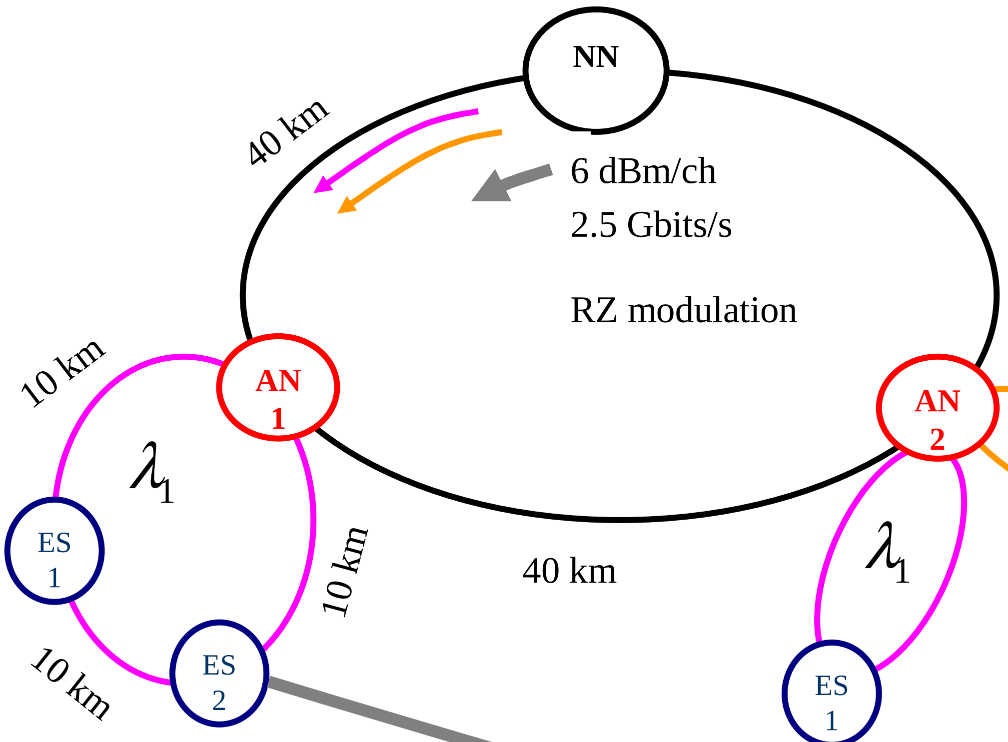




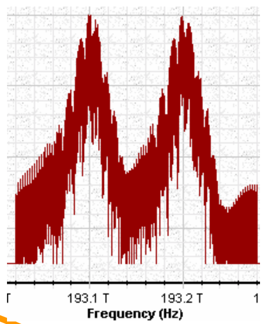
•P. P. Iannone et. al., "A flexible metro WDM ring using wavelength-independent subscriber equipment to share bandwidth", OFC'00, paper PD38, 2000.

•P. P. Iannone et. al., "A transparent WDM network featuring shared virtual rings", J. Lightwave Tech., vol. 18, pp 1955, 2000.

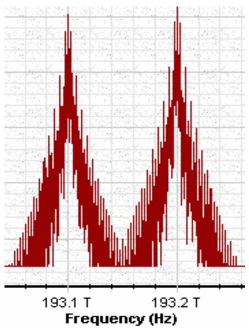
Simulation results for 2.5 Gbps



Directly modulated source

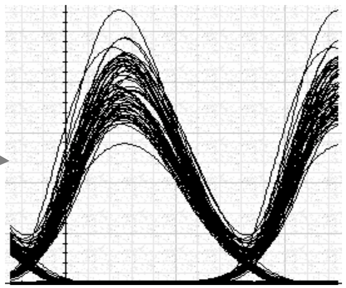


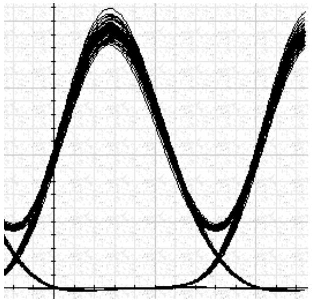
Externally modulated source



The line width of the source affects the broadening factor

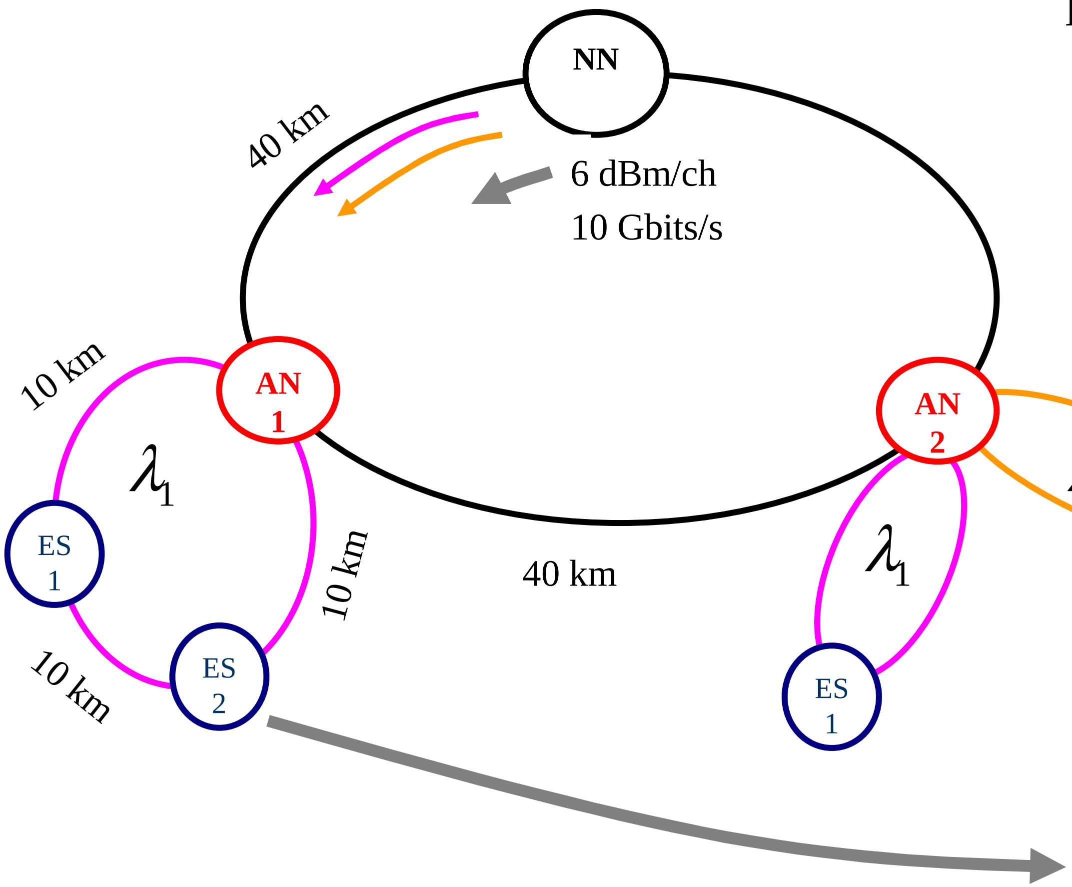
$$\frac{\sigma}{\sigma_0} = \left[\left(1+\frac{C\beta_2 z}{2\sigma_0^{2}}\right)^{2} + \left(1+[2\sigma_\omega\sigma_0]^{2}\right)\left(\frac{\beta_2 z}{2\sigma_0^{2}}\right)\right]$$



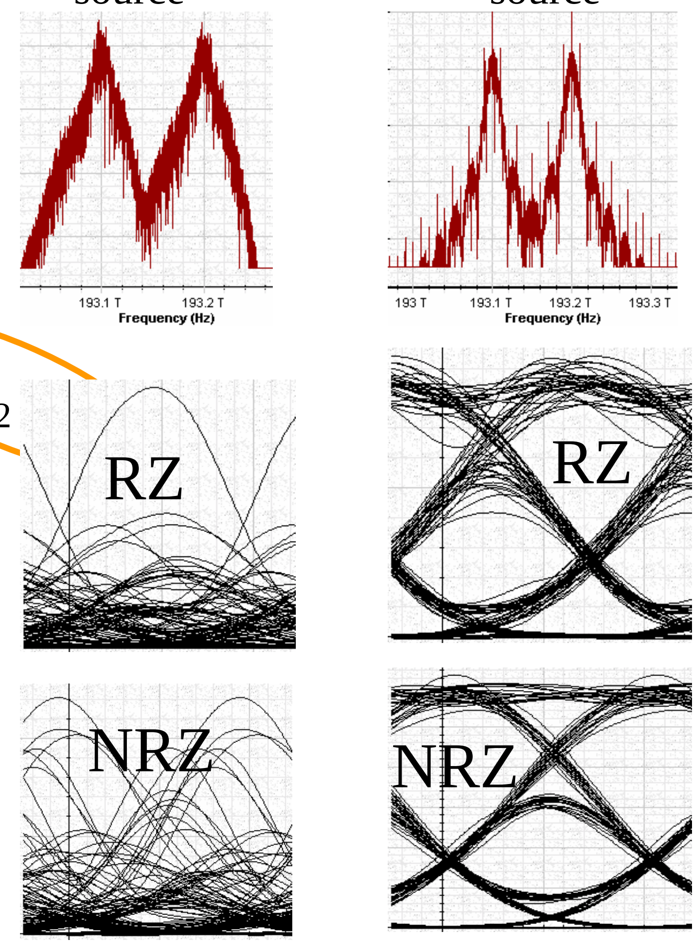


•G. P. Agrawal, Nonlinear Fiber Optics, Second Edition, Academic Press, 1995.

Simulation results for 10 Gbps



Directly modulated source | Externally modulated source

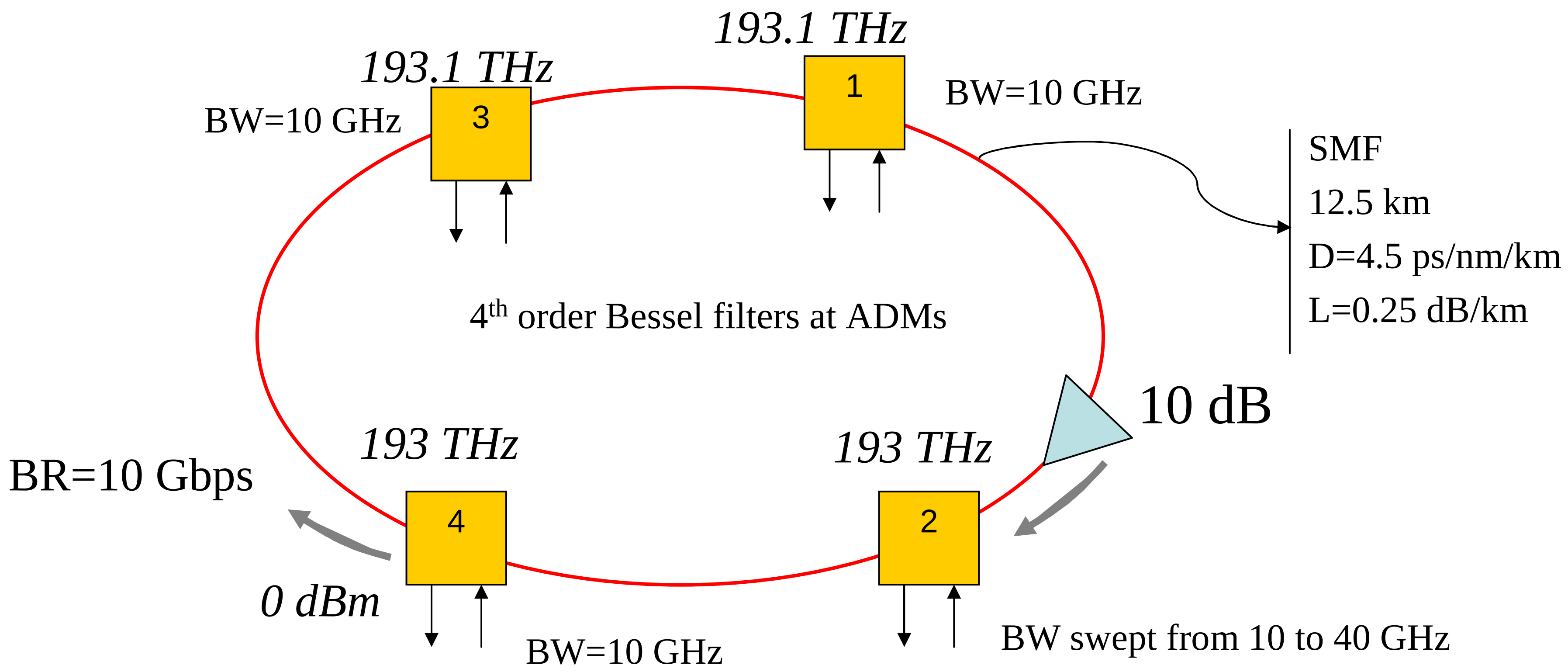


Dispersion compensation is required

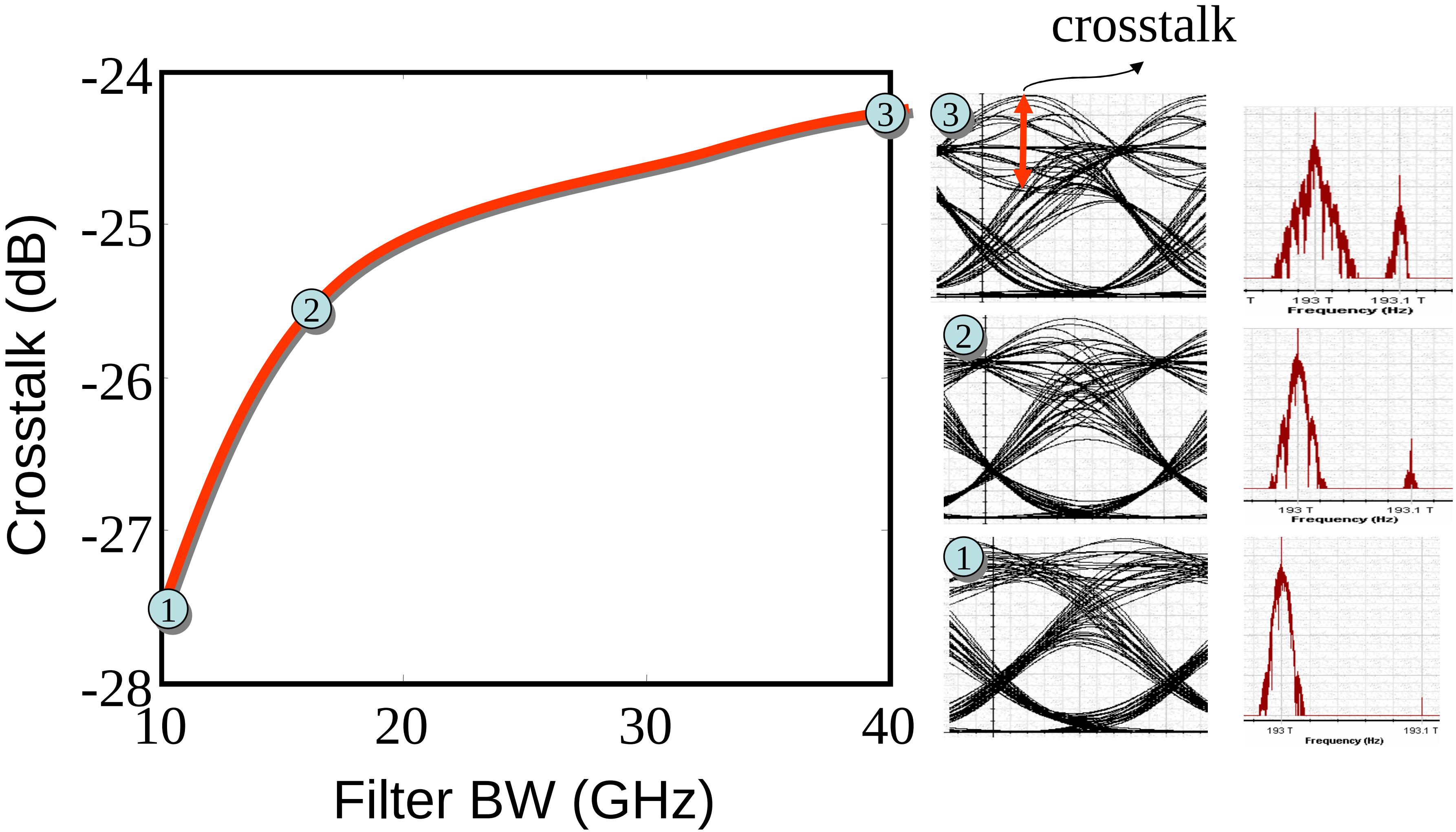
Inter-channel cross-talk

Goal of this example:

- *To simulate the inter-channel crosstalk at ADM and its effect to network performance*



Simulation results



•Tim Gyselings et. al, “Crosstalk analysis of multi-wavelength optical cross connects”, J. Light. Tech. 17, pp. 1273, 1999.

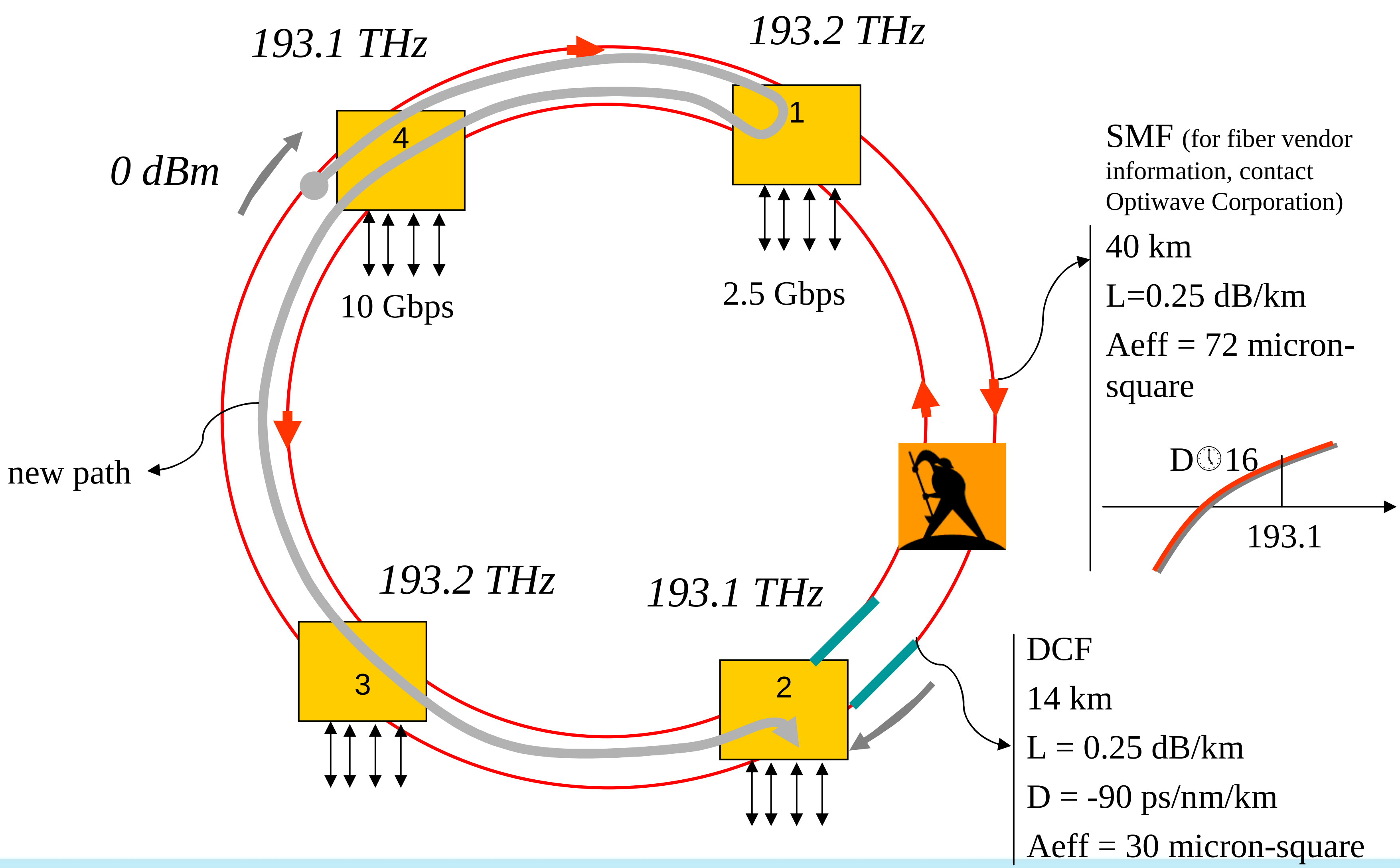
Dual fiber protected ring architecture

Goals of this example:

- *To show dual fiber protected ring architecture*
- *To show how a "lumped" dispersion compensation behaves for a dynamic network*
- *To show the effect of bit rate to the network performance*

•Peeya Iwagoshi, "High-speed ethernet optical architecture for metropolitan area networks", NFOEC'2001, 2001.

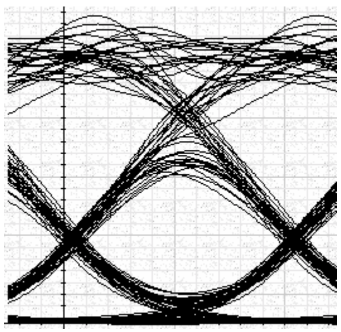
Dual fiber protected ring architecture



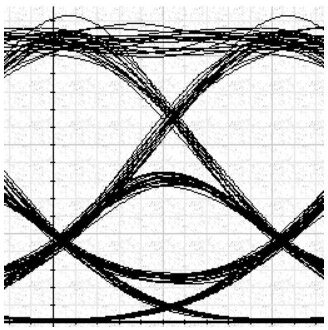
Simulation results

normal operation

fiber cut

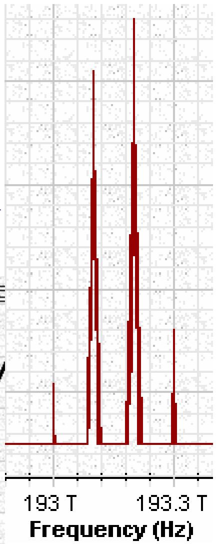


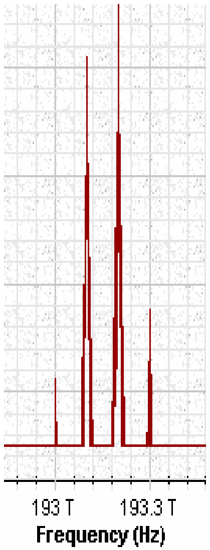
Ch 1 on primary path

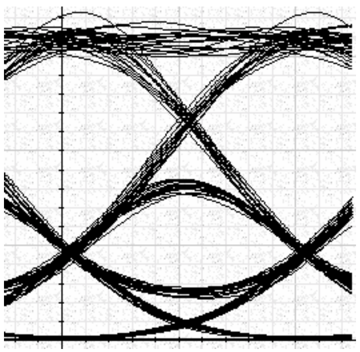


Ch 1 on secondary path

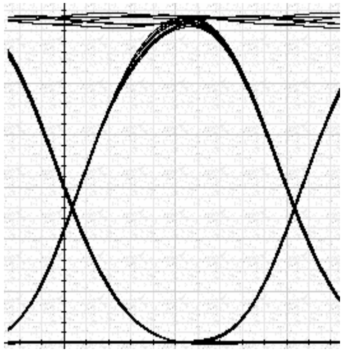
Spectrums over the paths



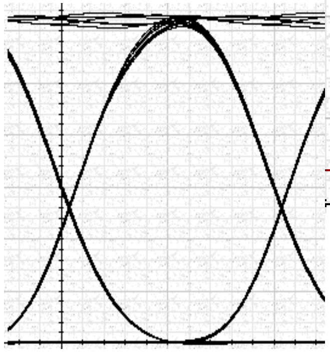




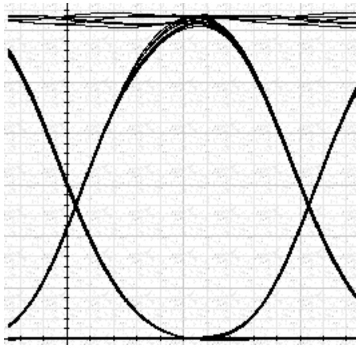
Ch 1 on new path



Ch 2 on primary path



Ch 2 on secondary path



Ch 2 on new path

Summary

- Key issues such as optical transparency, scalability, survivability, and low cost
- For power budgeting considering non ideal characteristics of amplifiers
- Dispersion at high bit rates and nonlinearity of fibers at high power levels
- Externally modulated and directly modulated light sources
- New fibers for metro applications
- Inter-channel cross-talk at OXCs
- Path recovery and influence to network performance
- Implementing new ideas